LAKESHORE STATE UNIVERSITY
Department of Statistical Science
STAT 3120: Applied Regression Analysis -- Fall 2026
Lectures: Tuesday/Thursday 10:10-11:25 am, Halvorsen Hall 140
Labs: Fridays, sections 201-204 (see the course site)
Course staff: one instructor and three teaching assistants. Office hours are posted on the course site.

1. COURSE DESCRIPTION
Linear models for real data: simple and multiple regression, inference for coefficients, diagnostics, transformations, model selection, logistic regression and an introduction to mixed models. Every topic is taught through a data analysis in R.

Prerequisite: STAT 2100 or an equivalent introductory statistics course.

2. TEXTS
- Course notes (free PDF on the course site), referred to below as "the Notes".
- Optional: any recent edition of a standard regression textbook.

3. GRADING
Problem sets (7, lowest dropped) 25%
Midterm 1 15%
Midterm 2 20%
Final project (proposal 5%, report 15%) 20%
Final exam 20%

4. PROBLEM SETS
Problem sets are posted on Monday and are due on Gradebox by 11:59 pm on the Friday listed. Late work loses 10% per day, up to three days.

Problem Set 1 (simple regression) -- due Friday, September 11
Problem Set 2 (inference for slopes) -- due Friday, September 25
Problem Set 3 (multiple regression) -- due Friday, October 9
Problem Set 4 (diagnostics and residual plots) -- due Friday, October 23
Problem Set 5 (transformations and interactions) -- due Friday, October 30
Problem Set 6 (model selection) -- due Friday, November 13
Problem Set 7 (logistic regression) -- due Friday, December 4

5. EXAMS
Midterm 1 is in class on Thursday, October 1 and covers Notes chapters 1-4.
Midterm 2 is an evening exam on Thursday, November 5, 7:30-9:00 pm, in Halvorsen Hall 140. It covers Notes chapters 5-8. One handwritten formula sheet is allowed.
The final exam is cumulative: Wednesday, December 16, 2:00-4:30 pm (room to be announced by the registrar).

6. FINAL PROJECT
You will analyze a public data set of your choice in a team of two.
- Project proposal (one page: question, data source, planned model) due Monday, November 2 at 11:59 pm.
- Final project report (8-10 pages with code appendix) due Friday, December 11 at 11:59 pm.

7. READING
Read Notes chapters 1-2 before the first lab on September 4.
Before Midterm 2, reread Notes chapter 7 on influence and leverage; it is the most-missed topic

every year.

8. POLICIES
Collaboration on problem sets is encouraged, but every student writes up their own solutions.
Accommodations: please send your accommodation letter in the first two weeks of the term.